

# Modular Network Device Architecture

Study Notes: MPU, LPU, SFU, and Packet Forwarding

# 1 The Big Idea

A network device (router, switch, firewall) has to answer three questions:

1. How does it forward a packet from one interface to another?
2. What hardware makes that possible?
3. How does that hardware work together?

On a normal PC, one general-purpose CPU does everything: the OS, applications, and network traffic. A high-performance router cannot do that – too many packets per second would overwhelm a general-purpose CPU. So the work is split into two roles:

- **Control plane** – the brain: decides what should happen (routing decisions).
- **Data plane** – the muscle: actually moves packets, very fast.

*The control plane thinks; the data plane forwards.*

## 2 The Three Logical Planes

- **Control plane** : the hardware that implements it MPU + LPU's local management unit
- **Data plane** : the hardware that implements it LPU (FPE/ASIC) + SFU.
- **Monitoring plane** : the hardware that implements it Monitoring units or CMU

## 3 The Hardware Components

### 3.1 MPU – Main Processing Unit

A small computer inside the device: CPU + RAM + storage + a network operating system (VRP, for Huawei). It provides:

**Control plane functions:** protocol processing (OSPF, BGP, IS-IS, RIP, ARP-related control info), route calculation, forwarding control, service processing (VPN, QoS), service scheduling, traffic statistics, system security.

**Management plane functions:** system status monitoring, environment monitoring, log and alarm processing, system software loading, system upgrade.

**Route calculation** = “Which path is best?”

**Forwarding control** = “Give the forwarding hardware what it needs to actually forward packets along that path.”

### 3.2 LPU – Line Processing Unit

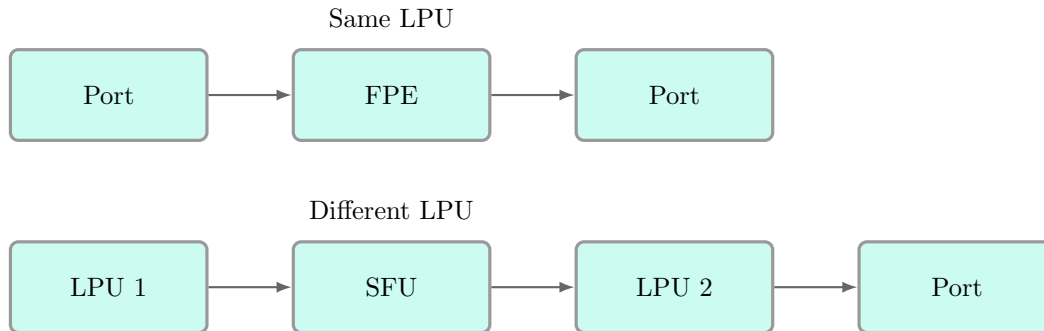
Where the physical ports live (GE, 10GE, etc.). Contains two different things that should not be mixed up:

- **LPU management unit** – talks to the MPU, local monitoring/control of the card. Part of the *control plane*.

- **FPE / ASIC / NP** – the high-speed forwarding chip. Part of the *data plane*. Handles packet forwarding, switching, QoS, statistics.

### 3.3 SFU – Switch Fabric Unit

The internal high-speed “highway” connecting different LPUs inside the chassis. Only needed when the destination port is on a *different* LPU.



### 3.4 ASIC, NP, and FPE – how they relate

- **ASIC** = a broad category: any chip built for one specialized job (as opposed to a general-purpose CPU).
- **NP (Network Processor)** = a specialized processor built specifically for networking tasks.
- **FPE** = the vendor’s name (Huawei) for the forwarding chip that lives inside the LPU.

FPE, ASIC, and NP are *functionally equivalent in role* (all do high-speed packet forwarding), but not guaranteed to be the exact same underlying technology – the real chip inside a given LPU might be implemented as an ASIC, an NP, or a hybrid.

## 4 Protocol Packets vs. Service Packets

Every packet a device receives is classified into one of two types:

| Type            | What it is                                                                          | Where it’s processed   |
|-----------------|-------------------------------------------------------------------------------------|------------------------|
| Protocol packet | Network-control traffic – ARP, OSPF, BGP, IS-IS, RIP, STP BP-DUs, ICMP echo request | Control plane (MPU)    |
| Service packet  | Ordinary user/data traffic passing through the device                               | Data plane (LPU + SFU) |

**The real test is not “is this traffic governed by a protocol.”** Everything is. The test is: is the device itself one of the two parties actually communicating, or is it just a middleman relaying traffic between two other devices?

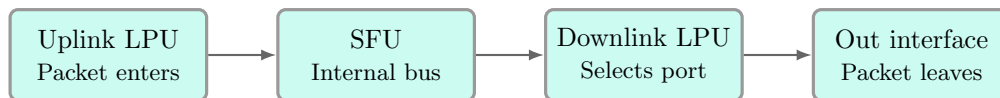
- OSPF/BGP/ARP packets are addressed *to the device itself* → protocol packets.
- HTTP/HTTPS/SMTP/FTP traffic between two end hosts is just being relayed → service packets, even though these are also technically “protocols.”

*Protocol packets are the network talking to itself; service packets are end-hosts talking through the network.*

#### 4.1 Protocol packet path (example: ARP)

1. An ARP packet arrives, recognized as a protocol packet, sent to the control plane.
2. The MPU makes two independent decisions: (1) should I respond? (2) should I learn the source MAC/IP?

#### 4.2 Service packet path



If the destination is on the *same* LPU the packet arrived on, the SFU is skipped entirely – the FPE forwards it locally.

#### 4.3 How the hardware recognizes protocol vs. service packets

The FPE/ASIC inspects fields as the packet arrives and matches them against pre-programmed rules (“punt” rules):

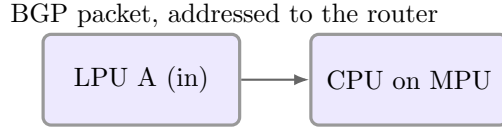
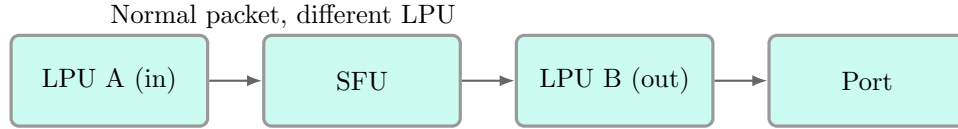
- **Destination IP/MAC is the device itself** → send to CPU.
- **Layer 2 header matches a known control protocol** (ARP, RARP, IS-IS, LLDP, LACP, PPP control) → send to CPU, no table lookup needed.
- **Destination IP is a reserved multicast address** (224.0.0.0 – 224.0.0.255, used by routing protocols) → send to CPU, no table lookup needed.

These rules are installed by the control plane at boot/initialization; the FPE just pattern-matches them at line rate – it is not making an intelligent decision, the MPU already told it what to look for.

#### 4.4 The trickier case: packets identified only by deep parsing (e.g. BGP)

BGP runs over ordinary unicast TCP (port 179), so it looks like normal traffic at a glance. It can only be identified by parsing deeper. Because of that, its path is described using the same generic model as an ordinary forwarded packet, with a special resolution:

- Destination board = the *same* board the packet arrived on (no real cross-chassis switching).
- “Outbound interface” = the CPU (treated as if it were a local port, but it is really the control plane).

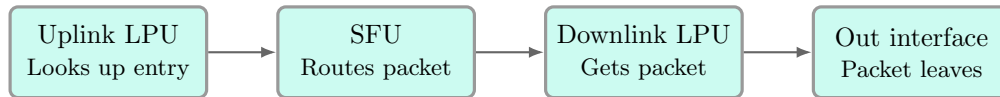


The BGP packet never crosses the SFU to reach a different board – it stays on the same inbound LPU and is handed directly up to the CPU. The trigger, in one line: **the LPU sends a packet to the CPU whenever the destination IP matches the device’s own address.**

## 5 Egress Determination and Forwarding Entry Storage

### 5.1 Where the lookup happens

When a packet enters the uplink LPU, the outbound interface and downlink LPU are determined *immediately*, by looking up the IP routing table / MAC address table – *before* the packet ever reaches the SFU. By the time the packet hits the SFU, the destination is already decided; the SFU does not do a lookup, it just switches based on what the uplink LPU already resolved.



### 5.2 Centralized vs. distributed forwarding entries

Where forwarding entries are stored is a trade-off between speed and resource usage.

- **Centralized:** the LPU asks the MPU for a decision on every packet. Simple, but the round-trip becomes a bottleneck at high packet rates.
- **Distributed:** the MPU still calculates routes, but pushes a local copy of the relevant entries down to every LPU. The LPU queries its own copy – much faster, at the cost of duplicated memory and synchronization overhead whenever a route changes.

Real high-performance devices use the distributed model: forwarding speed matters more than the extra memory cost.

## 6 Forwarding Entries, Forwarding Information, and Decoupling

### 6.1 Two different objects

- **Forwarding entries** – the master routing/MAC table the MPU builds and maintains, produced by running protocols and route calculation.
- **Forwarding information** – a separate, hardware-ready version derived *from* those entries, generated by the MPU and delivered down to the LPU. Not literally the same table – it has been converted into a form the LPU’s chip can use directly.

## 6.2 Forwarding-control decoupling

- **Control plane** (MPU): runs protocols, calculates routes, maintains forwarding entries, generates forwarding information, delivers it to the LPU.
- **Forwarding plane** (LPU / PFE): once it has that delivered information, it forwards packets completely independently – no per-packet trip to the MPU.

*If the MPU is heavily loaded, the forwarding plane is not affected – the two planes are decoupled.*

This is exactly why the distributed-entries model exists: the whole point of pushing forwarding information down to the LPU ahead of time is so the data plane never has to wait on the control plane during normal operation.

## 7 Processing of Protocol Packets in Detail

### 7.1 Why the MPU's CPU and the LPU's PFE are opposites

|                | General-purpose (flexible, slower)                            | Specialized (fixed job, very fast)               |
|----------------|---------------------------------------------------------------|--------------------------------------------------|
| PC             | CPU does everything                                           | —                                                |
| Network device | MPU's CPU – runs protocols, builds tables, constructs replies | PFE (NP/ASIC) on the LPU – only forwards packets |

Protocol packets need the MPU's CPU precisely because responding to ARP/OSPF/BGP requires flexible, general software logic (parsing, decision-making, building a custom reply) – something the fixed-function PFE cannot do. The PFE's only job is recognizing “this needs the CPU” and handing it off.

### 7.2 Rate limiting

The MPU's CPU has limited processing capacity. If too many protocol packets arrive, the CPU becomes busy and cannot respond in time. By default, the rate at which each type of protocol packet is sent to the CPU is limited, to protect it from being overwhelmed.

### 7.3 Sending a protocol packet back out

